

Assignment 02

1. Conditionals:
 - a. if-else [https://www.youtube.com/watch?v=MY03bt_0JQI],
 - b. switch statements
[<https://www.youtube.com/watch?v=Om3qzMoaIUo>].
 - c. ternary Operator
[<https://www.youtube.com/watch?v=kBdZBbWZ2a4>]
2. Loops:
 - a. for loops [<https://www.youtube.com/watch?v=vOmZ4JFhRds>]
 - b. While loops [<https://www.youtube.com/watch?v=ZjHJrmYknrk>]
 - c. Break & Continue
[<https://www.youtube.com/watch?v=iGo1Syv4YuM>]
 - d. Nested loops [<https://www.youtube.com/watch?v=sd4x2QG2K6U>]
3. Arrays
 - a. Need of arrays [<https://www.youtube.com/watch?v=uidBSlGLUK4>]
 - b. Creating array [<https://www.youtube.com/watch?v=239ubH043II>]

Project 02

Objective

Build a console application that handles multiple product orders in one flow while practicing Conditionals, Loops, and Arrays.

The Scenario

You are building a simplified checkout system for a shop.

Instead of handling one product, the shop owner now wants:

- multiple items in a single purchase
- automatic total calculation
- discount and delivery logic applied at the end

Requirements & Logic

Step 1: Ask Number of Items (Loop Control)

Ask the user: How many items do you want to purchase? (max 5)

- If user enters more than 5 → limit it using if-else

Step 2: Store Data (Arrays)

Create arrays:

- `String[] productNames`
- `double[] prices`
- `int[] quantities`

Step 3: Take Inputs (Loop)

Use a for loop to take input for each item:

- product name
- price
- quantity

Step 4: Input Validation (Conditionals + Continue)

Inside the loop:

- If $\text{price} \leq 0$ → show error and continue
- If $\text{quantity} \leq 0$ → show error and continue

Step 5: Calculate Totals (Loop)

For each item:

```
itemTotal = price * quantity
```

Store or directly accumulate:

```
subtotal += itemTotal
```

Step 6: Discount Logic (if-else)

Apply discount:

- if $\text{subtotal} > 100$ → 20%
- else if $\text{subtotal} > 50$ → 10%
- else → 5%

Step 7: Delivery Fee (Ternary Operator)

```
deliveryFee = (subtotal < 50) ? 10 : 0;
```

Step 8: Final Total

```
finalTotal = subtotal - discount + deliveryFee
```

Step 9: Final Receipt (Loop)

Print all items:

```
Item 1: Mouse - 2 x 500 = 1000
```

```
Item 2: Keyboard - 1 x 1500 = 1500
```

Then print:

- Subtotal
- Discount
- Delivery Fee
- Final Total

3. Expected Output

```
--- Welcome to BaseCase Shop ---
```

```
How many items do you want to purchase? 2
```

```
Enter Product 1 Name: Mouse
```

```
Enter Price: 500
```

```
Enter Quantity: 2
```

```
Enter Product 2 Name: Keyboard
```

```
Enter Price: 1500
```

```
Enter Quantity: 1
```

```
*****
```

```
--- Final Receipt ---
```

```
Item 1: Mouse - 2 x 500 = 1000
```

```
Item 2: Keyboard - 1 x 1500 = 1500
```

```
Subtotal: 2500
```

```
Discount Applied: 500
```

```
Delivery Fee: 0
```

```
Final Total: 2000
```

```
-----
```

[NOTE: You are allowed to apply improvements or any other techniques you want to. These are minimum requirements]

[NOTE: If you do not understand any part of the assessment or have any confusion feel free to ask right away]